The window for every offshore operation, decided.

For Brazil's 181 offshore production units: the next seven days of sea against your operation's limit — offloading, cargo, line laying, lifting, the Harbour Master's rule — over a forecast whose error was measured by satellite. Every decision comes out with its reason, and whoever checks it redoes it.

WHAT IS ALREADY MEASURED

0.33%

of the CLEARED windows with Hs ≤ 2 m that the sea broke, in 21,259 satellite comparisons the band never saw (since July 2025): **21 of 6,368**.

With a single provider's band (ECMWF), in the same test: 1.10%. The band that decides covers 96.5% of what the satellite saw; it claims 90%.

CLEARED — every limit holds at the measured band's unfavourable edge, at every step of the window

BLOCKED — a limit fails even at the favourable edge; it comes with the witness: the hour and the variable

UNDECIDED — the band straddles the limit; it comes with the threshold that would flip the decision

NEEDS DATA — the rule limits what nobody forecasts there (current, visibility), and says what

A forecast is a number; an operation needs a decision it can defend.

2.4 M FORECAST, 2.5 M LIMIT

A forecast without its error measured at the site does not say whether the window fits. Whoever decides rounds out of caution or out of haste — and a broken window can cost a vessel, while a missed one costs day rate and tank space.

DNV'S A CLOSES WINDOWS

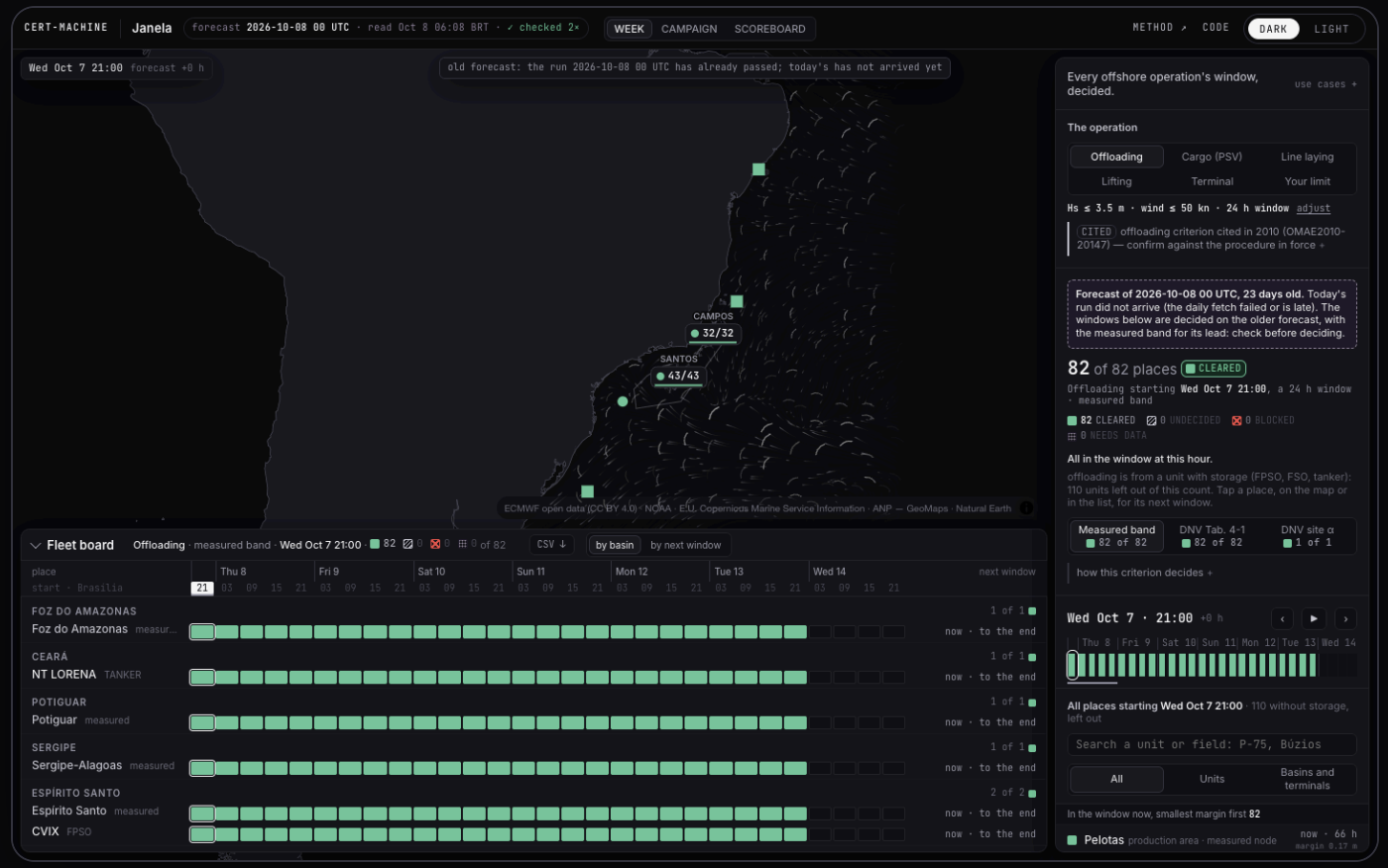
Table 4-1 (DNV-OS-H101), calibrated in the North Sea, shrinks the forecast limit. In Santos, Hs ≤ 2.0 m for 48 h: the sea opens **24.2%** of the starts; the table leaves **3.3%**.

THE PAPERWORK

Marine warranty, the insurer and HSE ask for the weather argument in writing. No provider publishes its own scoreboard against what the sea did.

> *“The expected uncertainty in the weather forecast should be calculated based on statistical data for the actual site and the operation schedule, i.e. TPOP.” — DNV-OS-H101, Section 4, B704. The standard already asks for the site's error; nobody measures it.*

The fleet on the map, the answer first.



The answer before the chart

How many units can start the operation at the chosen hour, and which cannot, by name.

The fleet board

Every unit against the week's 29 starts, by basin or by smallest margin, linked to the map; it exports as a spreadsheet (CSV).

Four verdicts, always with the word

Shape and word, never colour alone: CLEARED, BLOCKED with the witness, UNDECIDED with the threshold, NEEDS DATA with what is missing.

Three criteria side by side

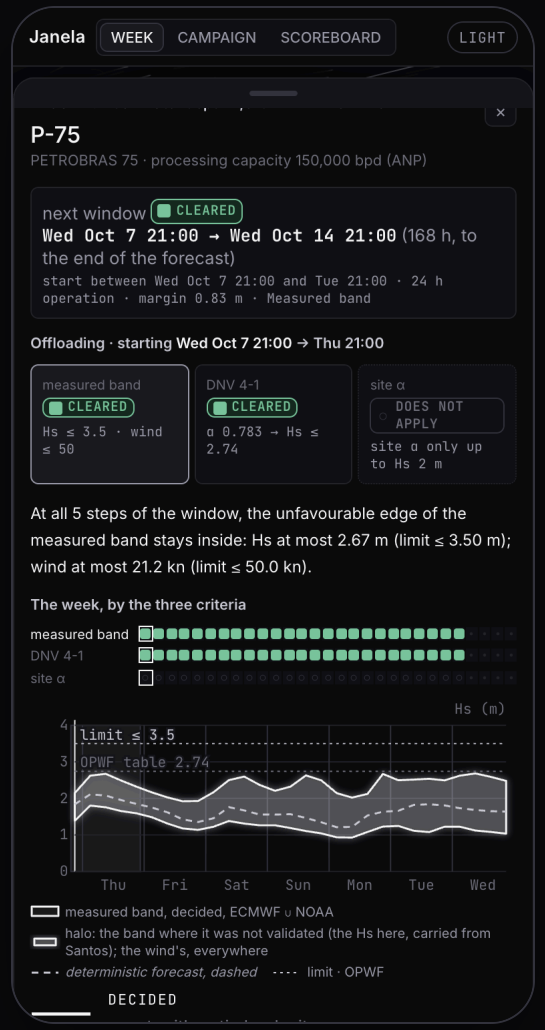
The measured band, DNV's Table 4-1 and the site's α, for the same window.

Redone in your browser

The day's 67,744 published decisions, re-decided in the tab by the same pinned modules, in seconds, and compared with the record.

Six decisions, and each one's door into the app.

WHO	THE QUESTION	WHAT THEY LEAVE WITH	HORIZON
marine operations coordination	Can this FPSO's offloading start, and when?	the verdict over the measured band, the margin to the limit, the threshold that would flip the decision and, with your tank numbers, the CRITICAL OFFLOADING warning	0–72 h
offshore logistics	Which unit does the PSV serve first this week?	the units by their next window, the smallest margin first, each with the week in 29 starts	1–7 days
campaign engineering	How many days do ten operations take from November on?	the campaign run over 32 years of past sea: the sea, Table 4-1 and the site's α, in vessel days	months
marine warranty survey and HSE	Does this weather argument hold?	the decision note and the .json certificate: inputs pinned by sha256, the rule cited, the commands to redo it by hand	per operation
management	Where is the weather costing us this week?	the units outside the window by name and, with your day rate, the cost of waiting for the next CLEARED window	the fleet
whoever doubts the forecast	How far off is this band?	the scoreboard: every band committed before the sea and scored by satellite, in public, with the admission rule written beforehand	the record



Every verdict with its reason, its margin and the paperwork to redo it.

The next window and its margin

When to start, until when, and how far the unfavourable edge sits from the limit — the margin rounded down, never overstated.

CRITICAL OFFLOADING, with your numbers

Storage, inventory and production typed in stay in the browser; Janela puts “tanks full in X h” beside the next offloading window and warns when it opens too late.

The cost of waiting

At your day rate — Janela never invents a day rate — until the next CLEARED window.

Decision note and .json certificate

The rule cited with its act and page, the run, the band, the sha256 of every module and record, and the commands to redo it by hand.

The forecast with its error measured by satellite, per site and lead time.

THE ARCHIVE

45,987

ECMWF forecast × satellite altimeter pairs at 8 sites, 2023–2026; plus 53,538 from NOAA WAVEWATCH III. A band calibrated per site and lead time, claiming 9 in 10.

OUT OF SAMPLE

96.5%

coverage in 21,259 comparisons since July 2025, which no band saw in calibration.

THE DECISION THAT COUNTS

0.33%

of the CLEARED windows with Hs ≤ 2 m broken by the sea (21 of 6,368); with ECMWF alone, 1.10% (87 of 7,900).

Two providers, ECMWF and NOAA: the **union** of the two calibrated bands decides, and it covers whenever either one covers — chosen by measurement, before deciding. The raw ensembles are shown and scored; they never decide.

A third, ECMWF's AI model (AIFS, open data since May 2026), was measured by the same rule, written beforehand: in 3,108 comparisons after 2026-08-01, the union IFS ∪ AIFS cleared 694 with 1 broken; IFS ∪ NOAA, 677 with 1. No measured difference where it matters first — it is scored every day and does not decide yet.

Over the offshore basins, the measured α sits above the North Sea table in every cell.

THE METHOD, CALIBRATED FIRST

16 of 20

Table 4-1 cells reproduced (±0.02) by the method that made the table, before looking at Brazil.

DESIGN HS 2 M · 7 OFFSHORE SITES

0.79–0.96

the site's α, against 0.63–0.76 in the table; the 90% interval sits above the table in 35 of 35 cells.

SANTOS · HS ≤ 2.0 M · 48 H

the sea allows	24.2%
Table 4-1 (α 0.73)	3.3%
the site's α (0.86)	10.8%

of operation starts, 1993–2024, counted exactly.

A statistical estimate (90% interval, resampled by day), used at its lower bound; the standard's 1-in-10,000 tail is a model extrapolation. On the coast (Florianópolis) and in Sergipe-Alagoas the site's α sits at or below the table — and Janela says that too. Evidence for the surveyor to redo, never an approval.

Ten 48 h line-laying operations in Santos from 1 November: how many vessel days?

CRITERION	HALF THE YEARS	9 IN 10	THE WORST YEAR
the sea (Hs ≤ 2.0 m)	**56.0 days**	74.8	92.6 (2013)
the site's α (OPWF 1.72 m)	**92.9 days**	140.6	152.1 (2003)
Table 4-1 (OPWF 1.46 m)	**226.9 days**	524.4	631.1 (2015)

Each of the 32 hindcast years runs the campaign: the first window after 1 November, the operation takes the window, the next one is sought from its end. The quantiles are exact order statistics of the years that finish (1 year in the row “the site's α” and 1 year in the row “Table 4-1” did not finish before the record ends: counted, left out). **The distance between the table and the site's α, in vessel days, is the argument the surveyor weighs.**

No transit or mobilisation wait; a hindcast (Ifremer WAVEWATCH III) is a model's past sea, and the count is exact over it — and the hindcast was checked against 14 deployments of the Brazilian Navy's buoys (PNBOIA, 2009–2018, 25,215 pairs): time with Hs ≤ 2.0 m, buoys 65.5%, hindcast 64.1%; below 1.5 m the hindcast is conservative (26% against 35%). In the app: any area, limit, window, number of operations and month.

Trust you can check, not trust you are asked for.

Exact arithmetic

Every verdict is decided in rationals on the limit as printed; the published band is rounded outward, so CLEARED and BLOCKED hold on the exact one.

A second checker

Written from the specification without reading the first (Python, no code in common): 67,744 of 67,744 decisions equal. The build refuses a day on which the two disagree.

Pinned inputs

The forecast, the bands, the Harbour Masters' local rules (NPCPs) and DNV's table enter by sha256; the decision note says which commit made the day and how to redo it.

Redone in the reader's browser

The modules that decide ship inside the page, pinned by sha256; the tab redoes every published decision of the day and compares it with the record.

A public scoreboard

42,403 bands committed before the target hour since 2026-10-06, scored by satellite; the rule that prunes a band was dated before the first datum. The record only grows.

What is said, and what is not

The band is decided at the forecast steps, at the model node — not the sea between steps, not the berth, not a probability.

The whole Brazilian margin, on data free for commercial use.

WHERE

181

ANP offshore production units (FPSO, FSO, platforms), each with the band of the nearest measured site; 9 sites measured against satellite, from Pelotas to Foz do Amazonas.

THE RULES

60

published limits from 11 acts of the Maritime Authority (NPCPs, the Harbour Masters' local rules, and ordinances), with page and literal quote; the offloading criterion cited (OMAE2010-20147); DNV's Table 4-1.

THE DATA

ECMWF open data (CC BY 4.0) · NOAA WAVEWATCH III (public domain) · NOAA altimetry (public domain) · Copernicus Marine currents (shown, not yet decided) · Ifremer hindcast (CC BY-SA 4.0) · ANP GeoMaps.

Collected automatically every day; each day published with the commit that made it. Janela runs in the browser: no number you type (inventory, day rate, limit) leaves it.

The limits, stated before anyone asks.

- **It is not an approval of an operation.** Marine warranty and the class societies own that word; Janela is evidence a surveyor redoes.
- **The model node, not the berth.** At a terminal, the forecast is the open sea of the approach: wave limits at the berth stay NEEDS DATA until a coastal transfer exists.
- **Current and visibility do not decide.** The current is shown and is being measured against observation; a rule that limits it stays NEEDS DATA, and says what would close it.
- **The site's α is an estimate;** the standard's 1-in-10,000 tail is an extrapolation. In Sergipe-Alagoas the band tends to be conservative (the satellite sees the sea 40–100 km off the coast).
- **The forward scoreboard is just starting:** the first bands are scored three days after each day closes; the first admission reading comes with 30 trial days.
- **The cargo, line-laying and lifting limits are examples** and say so; the offloading one is cited. Use your procedure's.

A pilot on your operations, judged by its own scoreboard.

WE ASK FOR

- one operation (offloading, PSV cargo, a campaign)
- your procedure's limits
- the units and an operations contact
- if there is one, the weather-downtime log, to compare

WE DELIVER

- the daily windows for those units, with a decision note
- the decisions checked by satellite, on the scoreboard
- the site's α argument, for the surveyor to redo
- all of it re-runnable, with the code open

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Re-runnable evidence about a window — not an approval of an operation nor a class certification; it does not replace the procedure or the responsibility of whoever signs. The numbers in this document are read from the repository's records at the moment it was generated.